1. Clinical Question & Core Hypothesis

Clinical Question

The ontological path links EPB41L2-driven cytoskeletal/adhesion programs to an ECM-remodeling/inflammatory output that prognostically matters, yet FGFR3-associated biology co-occurs with signals of better outcome. Why do stroma-rich, EPB41L2-associated tumors sometimes progress while others do not — could FGFR3 status modulate the deleterious stromal/ECM program? 2) Tier 1 path used: the single provided path (EPB41L2 → underlying risk → Osteopontin → prognosis → FGFR3) is my foundation because it directly connects adhesion/cytoskeletal perturbation to an ECM/inflammatory effector and a countervailing FGFR3-related favorable signature. 3) Tier 2/supporting evidence: empirical links between stromal remodeling and progression, and known roles of FGFR signalling in promoting differentiation/less invasive states, support the mechanistic plausibility that FGFR3 could blunt ECM-driven invasion; osteopontin (SPP1) is treated as the principal mediator and placed in secondary_modifiers. 4) Why PFI: the chain implicates ECM remodeling/invasion and thus risk of progression/recurrence more directly than late mortality, so progression-free interval (PFI) is the most targeted clinical endpoint to test the mechanistic question. 5) Deliberate exclusions: I excluded imaging features tied to immune organization (TIL density, TLS) because the path centers on ECM/cytoskeletal programs; I excluded naming osteopontin in the core hypothesis to keep the core variables to the allowed maximum (osteopontin/SPP1 is included as a secondary_modifier for mechanistic follow-up). I also excluded other ECM genes not present in the provided path to avoid inventing unsupported anchors?

Working Hypothesis

In tumors with a high stromal-to-tumor ratio (TSR), elevated EPB41L2 expression drives an ECM remodeling and inflammatory program that accelerates tumor progression, resulting in a shorter progression-free interval (PFI). This adverse TSR/EPB41L2 effect on PFI is mitigated when FGFR3 signaling is high, which promotes a more differentiated, less invasive tumor cell state and counteracts the ECM-driven progression program. The impact of the TSR/EPB41L2 axis on PFI is further modulated by SPP1 (osteopontin) signaling as a secondary modifier, such that higher SPP1 amplifies progression risk in the TSR/EPB41L2-high, FGFR3-low context. This framework yields a testable mechanistic axis linking imaging features (TSR) with the cytoskeletal regulator EPB41L2, ECM remodeling, and FGFR3-mediated differentiation, with PFI as the primary endpoint.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	EPB41L2, FGFR3
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

EPB41L2: EPB41L2-associated tumors sometimes progress while others do not — could FGFR3 status modulate the deleterious stromal/ECM program? 2) Tier 1 path used: the single provided path (EPB41L2 → underlying risk → Osteopontin → prognosis → FGFR3) is my foundation because it directly connects adhesion/cytoskeletal perturbation to an ECM/inflammatory effector and a countervailing FGFR3-related favorable signature.

FGFR3: FGFR3 status modulate the deleterious stromal/ECM program? 2) Tier 1 path used: the single provided path (EPB41L2 → underlying risk → Osteopontin → prognosis → FGFR3) is my foundation because it directly connects adhesion/cytoskeletal perturbation to an ECM/inflammatory effector and a countervailing FGFR3-related favorable signature.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	SPP1	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- SPP1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.